FROM YOU

YOUR SUGGESTIONS AND THINGS YOU WANTED TO KNOW!

EV CHARGING CIRCUIT

I need battery charging and DC converter circuits that are used in an electric vehicle (EV). Earlier I sent an email to you regarding this, but it seems you have not published any article on EV circuits yet.

Ravi Sankar

EFY. Please check 'Design Your Own Electric Vehicle Battery Charging Solutions' available on EFY website at *https://www.electronicsforu.com/electronics-projects/electronics-design-guides/electric-vehicle-battery-charging-solutions*

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PROJECTS WITH PCBS

I want EFY magazine issue that has projects with printed circuit boards (PCBs).

Yashu

EFY. There are DIY projects with PCB layouts in almost every issue of EFY magazine. Some of the projects with PCBs are also available on EFY website at *https://electronicsforu.com*

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ARDUINO BASED PROJECTS

I am looking for Arduino based projects. Please let me know some of the latest Arduino based projects published in EFY.

Karthi

EFY. Some of the latest Arduino based projects published in EFY are:

1. Weather Station With Device Control Using Touch Screen Display (April issue)
2. Study Of Rectified Waveforms Using Arduino Uno (March issue)
3. Smart Switch Box For Electric Iron (March issue)
4. Bluetooth Controlled Data Logger Robot For Soil Testing (February issue)
5. Contactless Smart Bin (December 2021 issue)
6. Automatic School Bell That Also Announces Alert Messages (December 2021 issue)
7. Taxi Fare Meter (Dec. 2021 issue)

ARDUINO BOARD

How to upload source code in an Arduino board and work with the board? Where can I buy an Arduino board?

Ganesh

EFY. Getting started with Arduino board and uploading the code could be a little confusing for some beginners. Because getting all the details from one source is not always easy. To get started with Arduino board, first you would need to install drivers, Arduino software or Integrated Development Environment (IDE), and then upload the code (called sketch) to the Arduino board.

The step-by-step procedure to work with an Arduino board is given below. It is applicable for the PCs using Windows as the operating system (OS), including Windows 7, 8, and 10 versions. The steps are also applicable for Windows XP but some of the dialogue windows may be different.

Follow the steps below to start with Arduino board:

1. Install Arduino IDE in your PC. You can download the latest version of Arduino software from *https://www.arduino.cc/en/software*

2. Connect your Arduino board to PC using a USB cable and wait for Windows to begin USB driver installation process.

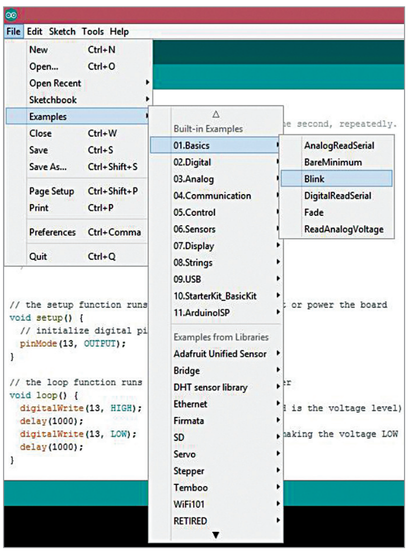

Fig. 1: Blink example from Arduino IDE

3. If the Windows installer does not start automatically, open the Windows Device Manager from Start→ Control Panel option and find the Arduino under Ports (COM&LPT) list.

4. If Arduino is not found, go to Other Devices and select the Unknown Device option. Then update the driver.

5. Select 'Browse my computer for driver software' option and go to the Arduino software download location and select Arduino.inf file/Arduino uno.inf (depending on your software version) to install the driver.

6. After successful installation of the driver, from the Arduino IDE select Tools/Board option. From here select your Arduino board among those listed, such as Arduino Uno, Arduino Mega 2560, Arduino Leonardo, etc.

7. Choose the correct serial COM port for your board. The COM port number will be visible under Device Manger.

8. Open the source code/sketch, compile it and upload the code to Arduino board by clicking the Upload button. If you do not have the sketch ready, the easiest way to start with the Arduino programming is to use the 'Blink' sketch from the Examples in Arduino. You can access Blink source code from File→Examples→Basics option, as shown in Fig. 1.

9. Compile the code and upload it to the Arduino board.

After uploading the code, you can access the Arduino pins or interface the pins with other devices. For example, the above Blink sketch is used to control (turn on/off) an LED through Arduino pin 13. You can change the LED blinking frequency by changing the code in Blink sketch.

Arduino boards—including Arduino Uno, Arduino Mega 2560, and Arduino Leonardo—are available from various online shops. You can also buy them from *www.kitsnspares.com* or contact them on *info@kitsnspares.com*

Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: **onedit@efy.in**)